

Feature Selection: Benchmark Hardness, Predictive Utility, and Resource Costs

From dataset taxonomy to preprocessing decisions

Dan Vilenchik

Ben-Gurion University of the Negev
Part of Itamar Elmakias's PhD thesis

Statistics Department, Hebrew University, June 2026

Last year: "Choosing the right dataset." This year: two follow-up questions about utility and resource allocation.

Recap: dataset hardness for feature-selection benchmarks

Feature selection (FS) is a standard pipeline step, with dozens of algorithms. We benchmark them on commonly used datasets — but **are those datasets hard?**

We built a hardness framework: run 27 FS algorithms on 102 datasets (63 binary, 39 multi-class) and label each dataset by the best AUC achievable.

- **Easy**: best AUC $\geq 90\%$
- **Medium**: 70–90%
- **Hard**: $< 70\%$

Recap: dataset hardness for feature-selection benchmarks

Feature selection (FS) is a standard pipeline step, with dozens of algorithms. We benchmark them on commonly used datasets — but **are those datasets hard?**

We built a hardness framework: run 27 FS algorithms on 102 datasets (63 binary, 39 multi-class) and label each dataset by the best AUC achievable.

- **Easy**: best AUC $\geq 90\%$
- **Medium**: 70–90%
- **Hard**: $< 70\%$

Hard 0

Medium 11

Easy 52

63 binary datasets.

No dataset is Hard under this protocol.

Published in *Knowledge-Based Systems* 334, 115022 (2026). doi.org/10.1016/j.knosys.2025.115022

High-dimensional theory predicts that hard instances should exist

Spiked covariance model (Johnstone; Baik–Ben Arous–Péché):

$$X_i = \sqrt{\lambda} z_i \mathbf{v} + \varepsilon_i, \quad \mathbf{v} \in \mathbb{R}^p, \quad z_i \sim \mathcal{N}(0, 1)$$

Recovering the signal $\mathbf{v}$ has a sharp phase transition at $\lambda_c = \sqrt{p/n}$:

- $\lambda < \lambda_c$: the leading sample eigenvector is asymptotically orthogonal to $\mathbf{v}$ — PCA carries no signal
- $\lambda > \lambda_c$: alignment emerges

Related high-dimensional recovery problems exhibit provable hard regimes (sparse PCA: Berthet–Rigollet; Krauthgamer–Nadler–V., *Ann. Stat.* 2015), which motivates expecting hard FS-like regimes.

High-dimensional theory predicts that hard instances should exist

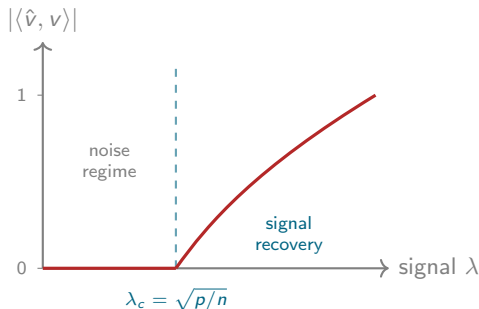
Spiked covariance model (Johnstone; Baik–Ben Arous–Péché):

$$X_i = \sqrt{\lambda} z_i \mathbf{v} + \varepsilon_i, \quad \mathbf{v} \in \mathbb{R}^p, \quad z_i \sim \mathcal{N}(0, 1)$$

Recovering the signal $\mathbf{v}$ has a sharp phase transition at $\lambda_c = \sqrt{p/n}$:

- $\lambda < \lambda_c$: the leading sample eigenvector is asymptotically orthogonal to $\mathbf{v}$ — PCA carries no signal
- $\lambda > \lambda_c$: alignment emerges

Related high-dimensional recovery problems exhibit provable hard regimes (sparse PCA: Berthet–Rigollet; Krauthgamer–Nadler–V., *Ann. Stat.* 2015), which motivates expecting hard FS-like regimes.



Below λ_c , even PCA's leading eigenvector is uninformative.

Empirical benchmarks often do not exhibit the predicted difficulty

Theory predicts: when p/n is large, FS should be unreliable. Many of our 102 datasets are high-dimensional ($p \gg n$).

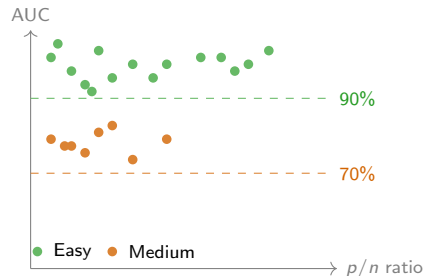
Empirical benchmarks often do not exhibit the predicted difficulty

Theory predicts: when p/n is large, FS should be unreliable. Many of our 102 datasets are high-dimensional ($p \gg n$).

What we observe:

- No significant correlation between p/n and performance
- No significant correlation between class imbalance and performance
- Many “high-dimensional” datasets are **Easy**

The hard regime exists in theory but is largely absent from the datasets commonly used for FS benchmarking.



No clear trend: high- p/n datasets are often Easy.

Generating hard benchmark variants by peeling

Since no naturally occurring Hard binary dataset was identified under the protocol, the published paper introduces **peeling**: shrink a dataset's training set while keeping a known ground-truth feature set F that still works.

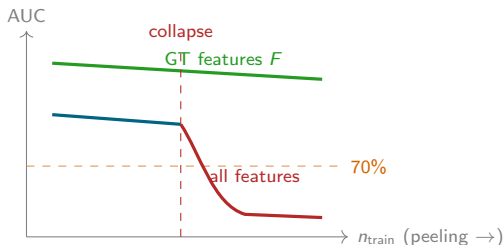
- Hard under our protocol: the 27 FS algorithms fail to recover F
- The retained set F still achieves good AUC, so the variant is not merely degraded noise
- Certified by construction, not a formal hardness theorem

Generating hard benchmark variants by peeling

Since no naturally occurring Hard binary dataset was identified under the protocol, the published paper introduces **peeling**: shrink a dataset's training set while keeping a known ground-truth feature set F that still works.

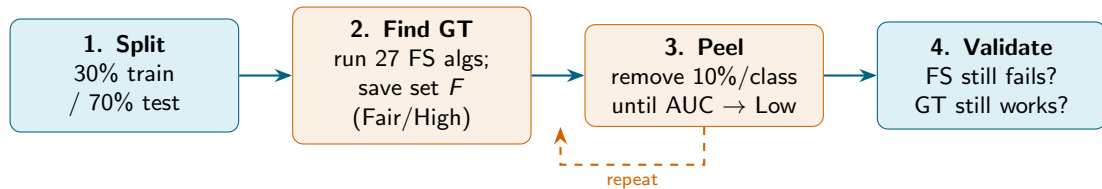
- Hard under our protocol: the 27 FS algorithms fail to recover F
- The retained set F still achieves good AUC, so the variant is not merely degraded noise
- Certified by construction, not a formal hardness theorem

Outcome: peeling all 63 binary datasets produced two Hard datasets — Slashdot and SMK-CAN-187.

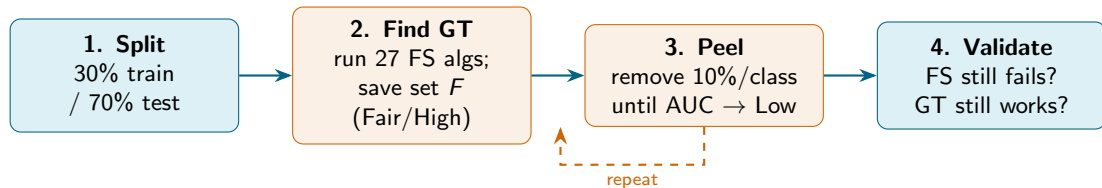


*A possible phase transition in FS recovery.
Procedure on the next slide.*

The peeling procedure



The peeling procedure



Stop when no-FS AUC drops to Low ($< 70\%$).

Fail if $n < 50$ or GT loses discriminative power.

Output: a dataset that is Hard under our protocol,
with a known F that still recovers good AUC.

Applied to all 63 binary datasets, only Slashdot and SMK-CAN-187 survived; both retain $> 80\%$ of samples.

From benchmark hardness to two follow-up questions

Last year we asked whether our benchmarks are hard. We now consider two follow-up questions.

Question 1

Can we predict whether FS will improve predictive performance?

Chapter 1 — utility prediction

From benchmark hardness to two follow-up questions

Last year we asked whether our benchmarks are hard. We now consider two follow-up questions.

Question 1

Can we predict whether FS will improve predictive performance?

Chapter 1 — utility prediction

Question 2

Conditional on using FS, which resource dimensions improve performance?

Chapter 2 — resource-aware evaluation

From benchmark hardness to two follow-up questions

Last year we asked whether our benchmarks are hard. We now consider two follow-up questions.

Question 1

Can we predict whether FS will improve predictive performance?

Chapter 1 — utility prediction

Question 2

Conditional on using FS, which resource dimensions improve performance?

Chapter 2 — resource-aware evaluation

Dataset hardness provides the organizing variable for both questions.

Question 1

Predicting feature-selection utility

Can dataset descriptors identify when FS improves predictive performance?

Prior work established that FS benefit is dataset-dependent

- **Surveys / benchmarks** show no universally best FS method; outcomes depend on the dataset and the learner (Bommert et al. 2020; Li et al. 2017).
- **Meta-learning** predicts FS choices, or FS benefit, from dataset descriptors (Filchenkov & Pendryak 2015; Parmezan et al. 2017).
- **Post et al. (2016)** is the closest prior work: it predicts whether CFS improves AUC for a *dataset* $\times$ *classifier* pair, across ~ 394 OpenML binary datasets and 12 classifiers, using a Random-Forest meta-model with FS landmarks.

Prior work established that FS benefit is dataset-dependent

- **Surveys / benchmarks** show no universally best FS method; outcomes depend on the dataset and the learner (Bommert et al. 2020; Li et al. 2017).
- **Meta-learning** predicts FS choices, or FS benefit, from dataset descriptors (Filchenkov & Pendryak 2015; Parmezan et al. 2017).
- **Post et al. (2016)** is the closest prior work: it predicts whether CFS improves AUC for a *dataset* $\times$ *classifier* pair, across ~ 394 OpenML binary datasets and 12 classifiers, using a Random-Forest meta-model with FS landmarks.

Our question is upstream and dataset-level: before running an FS search, can we identify datasets where any meaningful FS gain is likely?

Relation to Post et al. (2016)

Post et al. (2016) directly ask whether feature selection improves performance for a given dataset and classifier — the closest precedent to our question. We build on it and shift the deployment question.

Relation to Post et al. (2016)

Post et al. (2016) directly ask whether feature selection improves performance for a given dataset and classifier — the closest precedent to our question. We build on it and shift the deployment question.

	Post et al. (2016)	This work
Question	Does FS improve a specified classifier on this dataset?	Is an FS search worthwhile for this dataset?
Decision unit	dataset $\times$ classifier	dataset-level screening decision
FS intervention	mainly CFS subset selection	oracle over 37 FS algorithms / multiple subset sizes
Utility label	any / statistically significant improvement	meaningful gain, $\Delta\text{AUC} \geq 0.05$
Learning	supervised meta-learning over many labeled pairs	rare-event detection + weak-positive supervised transfer
Positives	41% of pairs improve, 10% significantly	7 / 102 datasets with substantial oracle gain

Relation to Post et al. (2016)

Post et al. (2016) directly ask whether feature selection improves performance for a given dataset and classifier — the closest precedent to our question. We build on it and shift the deployment question.

	Post et al. (2016)	This work
Question	Does FS improve a specified classifier on this dataset?	Is an FS search worthwhile for this dataset?
Decision unit	dataset $\times$ classifier	dataset-level screening decision
FS intervention	mainly CFS subset selection	oracle over 37 FS algorithms / multiple subset sizes
Utility label	any / statistically significant improvement	meaningful gain, $\Delta\text{AUC} \geq 0.05$
Learning	supervised meta-learning over many labeled pairs	rare-event detection + weak-positive supervised transfer
Positives	41% of pairs improve, 10% significantly	7 / 102 datasets with substantial oracle gain

We build directly on POST — FS utility is predictable. We ask a different deployment question: when is it worth initiating an FS search at all?

Oracle estimate of feature-selection utility

Setup: 37 FS algorithms $\times$ 102 datasets $\times$ 5 classifiers.

(This study expands Paper 1's 27 algorithms to 37.)

For each dataset $\mathcal{D}_i$,

$$\Delta\text{AUC}(\mathcal{D}_i) = \text{AUC}^{\text{FS}} - \text{AUC}^{\text{no-FS}},$$

where AUC^{FS} is the best result over all algorithms, classifiers, and feature counts K .

Threshold: $\Delta\text{AUC} \geq 0.05$ counts as a substantial gain.

Oracle estimate of feature-selection utility

Setup: 37 FS algorithms $\times$ 102 datasets $\times$ 5 classifiers.
(This study expands Paper 1's 27 algorithms to 37.)

For each dataset $\mathcal{D}_i$,

$$\Delta\text{AUC}(\mathcal{D}_i) = \text{AUC}^{\text{FS}} - \text{AUC}^{\text{no-FS}},$$

where AUC^{FS} is the best result over all algorithms, classifiers, and feature counts K .

Threshold: $\Delta\text{AUC} \geq 0.05$ counts as a substantial gain.

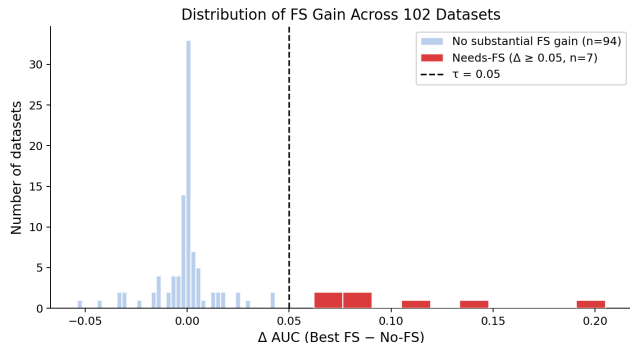
The oracle is deliberately generous — it selects, in hindsight, the best

- algorithm,
- number of features K ,
- classifier

per dataset.

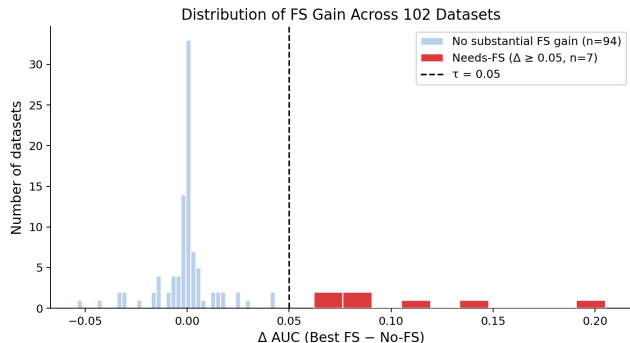
If gains are uncommon under this oracle evaluation, they are unlikely to be common under ordinary model-selection constraints.

Substantial predictive gains are rare in this benchmark



One of the 102 datasets is omitted from this histogram (legend totals 101).

Substantial predictive gains are rare in this benchmark



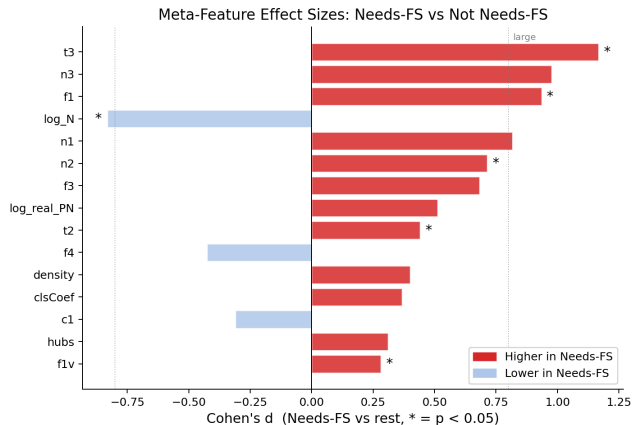
One of the 102 datasets is omitted from this histogram (legend totals 101).

Only **7 of 102** datasets (6.9%) reach $\Delta \text{AUC} \geq 0.05$ — even under the generous oracle.

- Most datasets cluster near $\Delta = 0$
- Some show lower AUC after FS
- The beneficial datasets lie in the upper tail of the empirical distribution

Those datasets account for under 1% of the total FS search cost; most FS evaluations are spent on datasets with little observed predictive gain.

Beneficial datasets differ in geometry, not raw dimensionality

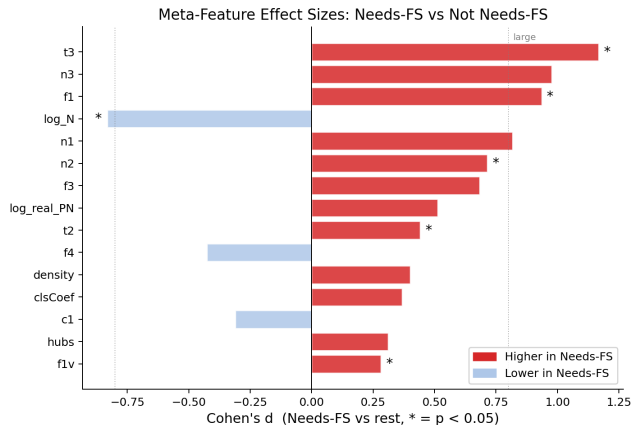


Effect sizes (Cohen's d), Needs-FS vs. the rest:

- Lower baseline AUC (more headroom)
- More class overlap
- More complex boundary geometry
- Smaller sample size

These descriptors are consistent with the difficulty descriptors used in the hardness taxonomy — not raw p , but problem geometry.

Beneficial datasets differ in geometry, not raw dimensionality



Effect sizes (Cohen's d), Needs-FS vs. the rest:

- Lower baseline AUC (more headroom)
- More class overlap
- More complex boundary geometry
- Smaller sample size

These descriptors are consistent with the difficulty descriptors used in the hardness taxonomy — not raw p , but problem geometry.

The separation is large enough that beneficial datasets form a distinct, rare population — motivating a **rare-event (anomaly) detection** approach.

One-class detection of beneficial datasets

One-class setup:

- Train on only the 95 No-FS datasets
- Test on 7 Needs-FS + 7 random No-FS
- Repeat 50 times

Models: PCA reconstruction, autoencoder, One-Class SVM, LOF, Isolation Forest.

No positive examples are used in training; the detector models the majority population and flags deviations.

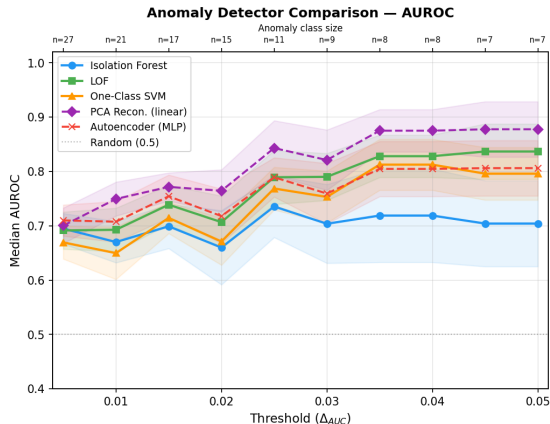
One-class detection of beneficial datasets

One-class setup:

- Train on only the 95 No-FS datasets
- Test on 7 Needs-FS + 7 random No-FS
- Repeat 50 times

Models: PCA reconstruction, autoencoder, One-Class SVM, LOF, Isolation Forest.

No positive examples are used in training; the detector models the majority population and flags deviations.



Unsupervised detection identifies most beneficial datasets

Model	AUROC	Recall
Autoencoder	0.90	0.71
PCA reconstruction	0.89	0.86
OC-SVM	0.89	0.86
LOF	0.87	0.71
Isolation Forest	0.80	0.29
Random baseline	0.50	—

Unsupervised detection identifies most beneficial datasets

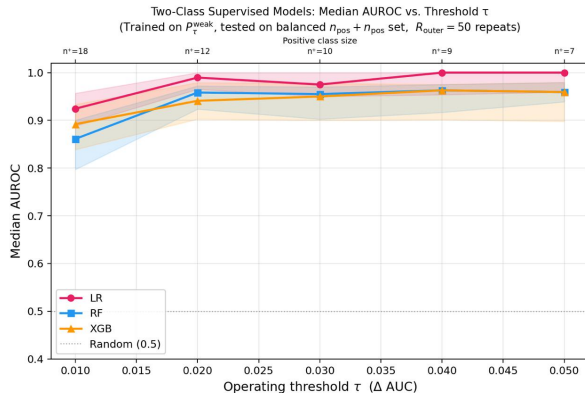
Model	AUROC	Recall
Autoencoder	0.90	0.71
PCA reconstruction	0.89	0.86
OC-SVM	0.89	0.86
LOF	0.87	0.71
Isolation Forest	0.80	0.29
Random baseline	0.50	—

With no positive training examples, simple detectors reach AUROC 0.87–0.90 in this evaluation.

Datasets assigned low anomaly scores may be candidates for bypassing exhaustive FS evaluation.

Recall is the primary concern, to avoid missing datasets that would benefit from FS.

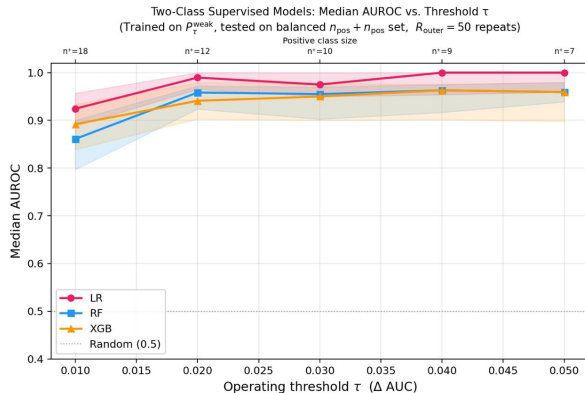
Weak positive examples improve supervised utility prediction



Use weakly beneficial datasets as positives:

- Train positives: $0.01 \leq \Delta \text{AUC} < 0.05$
- Test positives: $\Delta \text{AUC} \geq 0.05$ (held out)

Weak positive examples improve supervised utility prediction



Use weakly beneficial datasets as positives:

- Train positives: $0.01 \leq \Delta\text{AUC} < 0.05$
- Test positives: $\Delta\text{AUC} \geq 0.05$ (held out)

In this evaluation, logistic regression achieves AUROC 1.00; RF and XGBoost reach 0.96; no positive test cases are missed.

Weakly and strongly beneficial datasets share a structural signature that is approximately linearly separable in this representation.

Intrinsic descriptors outperform provenance labels

Does the model rely on dataset provenance (e.g. genomic data)?

Feature representation	AUROC
Intrinsic descriptors	1.00
Domain / provenance only	0.28
Combined	1.00

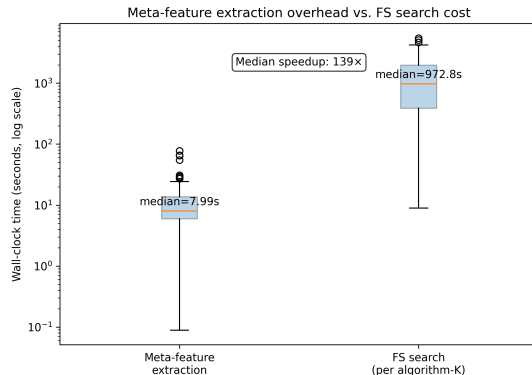
Domain labels alone are not predictive in this benchmark; intrinsic dataset descriptors carry the signal. Genomic data is 71% of Needs-FS and 75% of No-FS, so domain does not determine utility.

Intrinsic descriptors outperform provenance labels

Does the model rely on dataset provenance (e.g. genomic data)?

Feature representation	AUROC
Intrinsic descriptors	1.00
Domain / provenance only	0.28
Combined	1.00

Domain labels alone are not predictive in this benchmark; intrinsic dataset descriptors carry the signal. Genomic data is 71% of Needs-FS and 75% of No-FS, so domain does not determine utility.



Meta-feature screening has substantially lower median runtime than a single FS evaluation (≈ 8 s vs. ≈ 973 s; median ratio ≈ 139).

Question 2

Resource-aware evaluation of feature-selection methods

Which resource dimensions are associated with improved performance?

Prior work gives per-dataset winners, not transparent rules

- **Surveys / benchmarks** compare algorithms across datasets and report *dataset-specific winners* — no single method dominates (Bommert et al. 2019/2020).
- **Meta-learning** predicts the best FS method *per dataset* from meta-features (Filchenkov 2015; Parmezan et al. 2017): a separate single-point prediction for each dataset.
- **Complexity descriptors** (e.g. Proplexity) are high-dimensional and highly dataset-specific.

Prior work gives per-dataset winners, not transparent rules

- **Surveys / benchmarks** compare algorithms across datasets and report *dataset-specific winners* — no single method dominates (Bommert et al. 2019/2020).
- **Meta-learning** predicts the best FS method *per dataset* from meta-features (Filchenkov 2015; Parmezan et al. 2017): a separate single-point prediction for each dataset.
- **Complexity descriptors** (e.g. Proplexity) are high-dimensional and highly dataset-specific.

The gap

Per-dataset, single-point predictions can be accurate but are *fragmented*: this does not directly yield stable resource-level guidance, and it does not say whether spending *more* resources (runtime, supervision, custom code) is justified in the first place.

Prior work gives per-dataset winners, not transparent rules

- **Surveys / benchmarks** compare algorithms across datasets and report *dataset-specific winners* — no single method dominates (Bommert et al. 2019/2020).
- **Meta-learning** predicts the best FS method *per dataset* from meta-features (Filchenkov 2015; Parmezan et al. 2017): a separate single-point prediction for each dataset.
- **Complexity descriptors** (e.g. Proplexity) are high-dimensional and highly dataset-specific.

The gap

Per-dataset, single-point predictions can be accurate but are *fragmented*: this does not directly yield stable resource-level guidance, and it does not say whether spending *more* resources (runtime, supervision, custom code) is justified in the first place.

Our reframing: group methods into coarse resource categories and test, with paired within-dataset statistics, whether higher cost systematically pays — rules of thumb rather than per-dataset point estimates.

Grouping methods by resource cost

37 algorithms, 102 datasets, 5 classifiers.

Rather than ranking individual algorithms only, we group methods by resource cost and test whether higher-cost categories yield systematic gains.

Paired within-dataset comparisons: Wilcoxon signed-rank tests, win rates, Cliff's δ .

Grouping methods by resource cost

37 algorithms, 102 datasets, 5 classifiers.

Rather than ranking individual algorithms only, we group methods by resource cost and test whether higher-cost categories yield systematic gains.

Paired within-dataset comparisons: Wilcoxon signed-rank tests, win rates, Cliff's δ .

Five resource dimensions:

- ① Runtime (fast / moderate / slow)
- ② Implementation (library / research code)
- ③ Supervision (supervised / unsupervised)
- ④ Method family (filter / wrapper / embedded / ...)
- ⑤ Optimization effort (All vs. Best)

Two ways to use FS: “All” vs. “Best”

All (low effort)
pool every FS config

applying FS with-
out systematic tuning

Two ways to use FS: “All” vs. “Best”

All (low effort)
pool every FS config

applying FS without systematic tuning

Best (high effort)
keep best config per category

careful search within a fixed resource budget

Two ways to use FS: “All” vs. “Best”

All (low effort)
pool every FS config

applying FS without systematic tuning

Best (high effort)
keep best config per category

careful search within a fixed resource budget

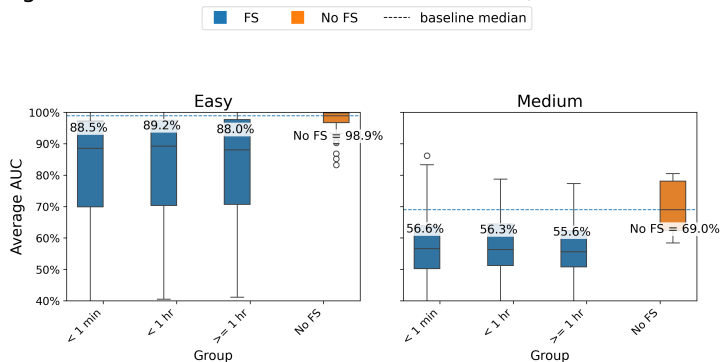
We also split results by hardness (Paper 1 taxonomy). Using the 37-algorithm suite, the binary split is **48 Easy / 15 Medium** (vs. 52 / 11 with 27 algorithms; the larger suite reclassifies a few datasets).

The distinction is important because aggregate conclusions differ across these two regimes.

Medium results should be interpreted cautiously: only 15 paired datasets, so a non-significant Medium comparison means “no evidence of a difference,” not “no difference.”

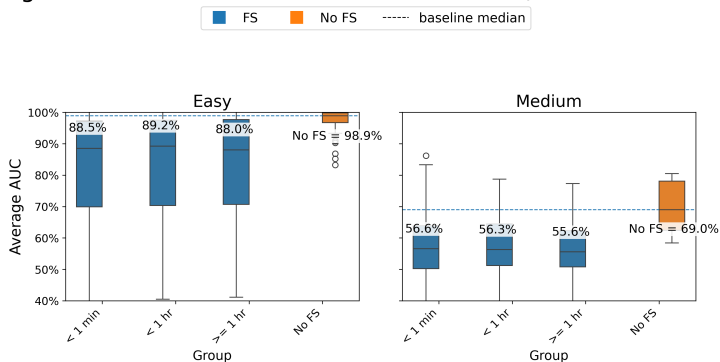
Untuned FS configurations underperform the no-FS baseline

Figure 1: Core · All · Performance · runtime (Runtime Bucket)



Untuned FS configurations underperform the no-FS baseline

Figure 1: Core · All · Performance · runtime (Runtime Bucket)



All setting: fast, moderate, and slow methods all land below the no-FS baseline.

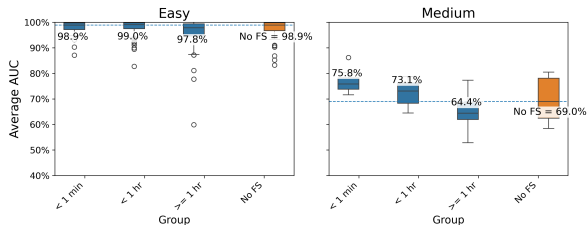
Easy: 88–89% vs. no-FS **98.9%**.

Medium: ~56% vs. no-FS **69.0%**.

Untuned FS lowers AUC by 7–12 points relative to no-FS. The dominant difference is between unselected configurations and the best configuration within a category.

Longer runtime does not yield systematic AUC gains

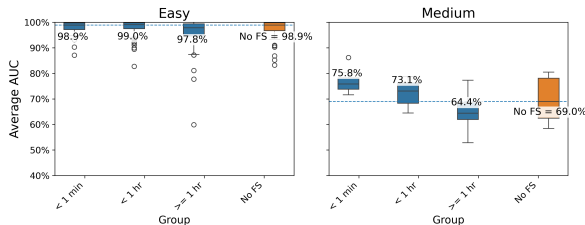
Figure 2: Core · Best · Performance · runtime (Runtime Bucket)



Core datasets, “Best” setting, by runtime bucket.

Longer runtime does not yield systematic AUC gains

Figure 2: Core · Best · Performance · runtime (Runtime Bucket)



Core datasets, “Best” setting, by runtime bucket.

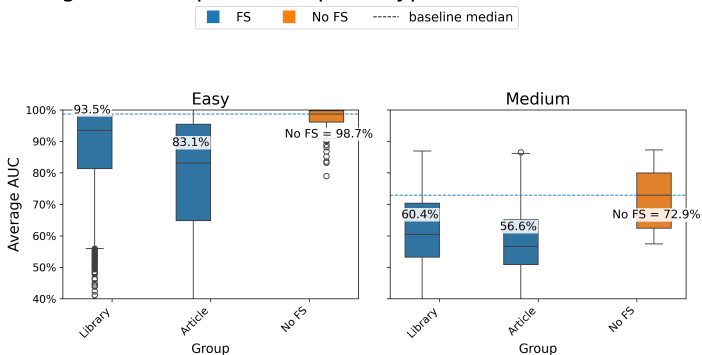
Slow methods do not outperform fast ones.

- Cliff's $\delta = 0.72$ on Easy (paired comparisons favor fast methods)
- Same direction on Medium, but with $n = 15$ the test is underpowered; we report it as a consistent trend, not a significant effect

On Easy datasets we find no evidence that higher runtime improves predictive performance; the direction holds across All and Best.

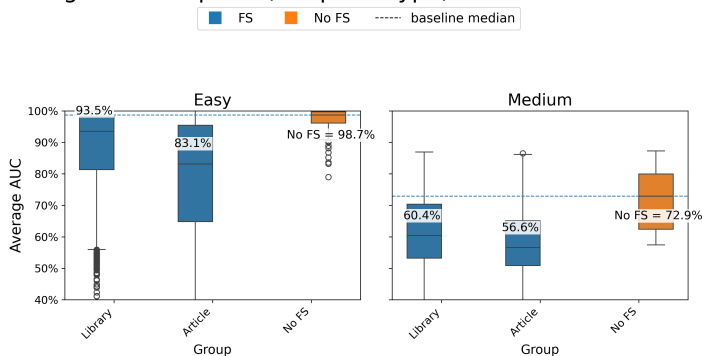
Library implementations are more robust in this benchmark

Figure 1: adoption (Adoption Type) · Performance · All



Library implementations are more robust in this benchmark

Figure 1: adoption (Adoption Type) · Performance · All



Easy (All): library 93.5% vs. research code 83.1%.

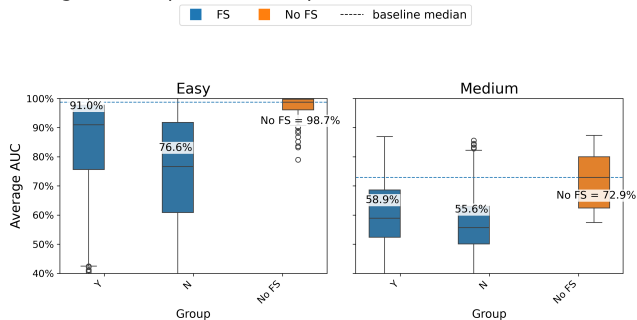
Medium (All): 60.4% vs. 56.6%.

The gap is larger on Medium datasets.

Research-code implementations are more sensitive to suboptimal settings; a plausible explanation is that maintained implementations have more stable defaults and broader testing.

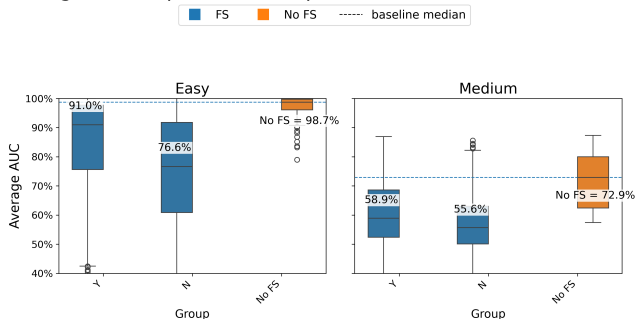
Supervision is the main positive resource effect

Figure 1: supervision (Supervised) · Performance · All



Supervision is the main positive resource effect

Figure 1: supervision (Supervised) · Performance · All

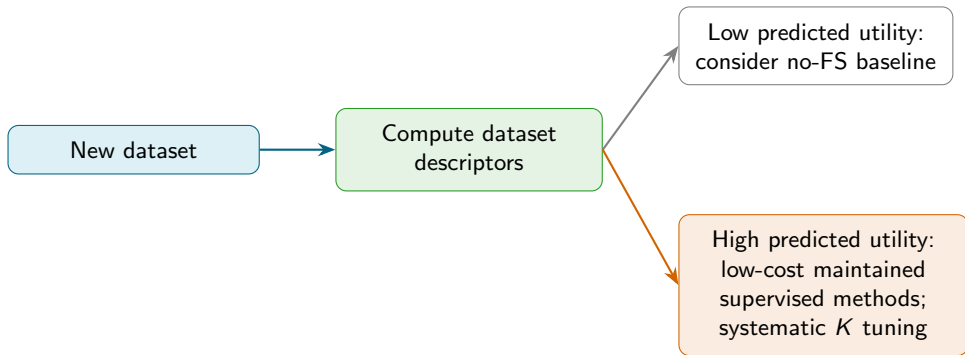


Easy (All): supervised 91% vs. unsupervised 76.6%.

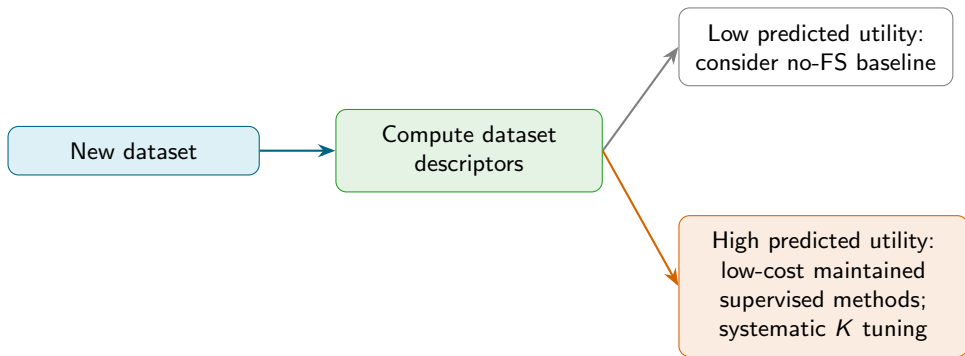
Medium: 58.9% vs. 55.6%.

Clear positive effect on Easy ($n = 48$, $p \approx 3 \times 10^{-8}$). On Medium it is small and only marginally significant ($n = 15$, $p \approx 0.028$) — consistent with both a weaker effect and limited power, so we do not over-interpret it.

Proposed evaluation workflow



Proposed evaluation workflow



Use dataset-level screening before exhaustive FS evaluation; begin with low-cost methods and systematic tuning; consider higher-cost methods only if the low-cost search is insufficient.

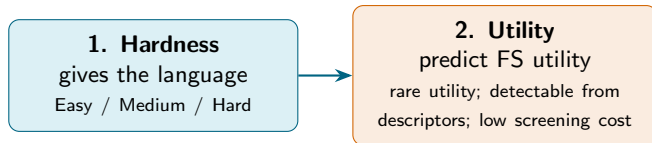
The decision pipeline

1. Hardness

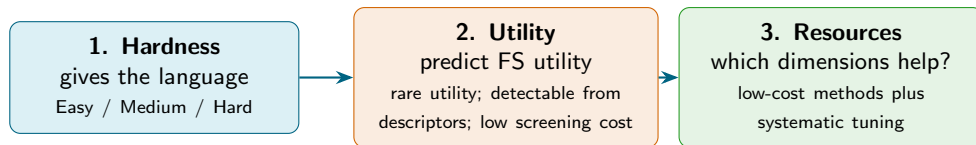
gives the language

Easy / Medium / Hard

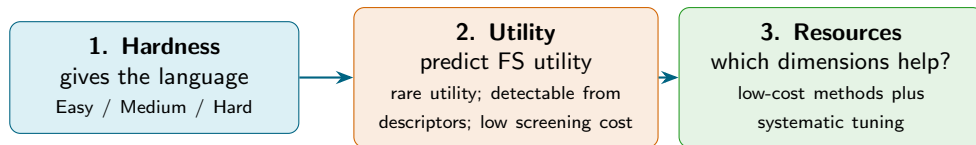
The decision pipeline



The decision pipeline



The decision pipeline



Dataset hardness provides a useful organizing variable for deciding whether FS is useful and how resource-intensive the evaluation should be.

Theoretical

- Is there a provable phase transition for feature recovery as n/p varies? The peeling collapse and the published Easy/Medium/Hard taxonomy are consistent with one.
- Can RMT / sparse-recovery tools characterize the FS-hard regime?

Open questions

Theoretical

- Is there a provable phase transition for feature recovery as n/p varies? The peeling collapse and the published Easy/Medium/Hard taxonomy are consistent with one.
- Can RMT / sparse-recovery tools characterize the FS-hard regime?

Empirical

- More Hard datasets — a systematic search?
- Does utility prediction transfer to new domains (time series, graphs, regression)?
- Can the utility predictor inform AutoML pipelines?

Open questions

Theoretical

- Is there a provable phase transition for feature recovery as n/p varies? The peeling collapse and the published Easy/Medium/Hard taxonomy are consistent with one.
- Can RMT / sparse-recovery tools characterize the FS-hard regime?

Empirical

- More Hard datasets — a systematic search?
- Does utility prediction transfer to new domains (time series, graphs, regression)?
- Can the utility predictor inform AutoML pipelines?

These results support treating FS as a conditional preprocessing step whose predictive value depends on measurable dataset properties.

Thank you

vilenchi@bgu.ac.il

Published paper: Itamar Elmakias & Dan Vilenchik, "Choosing the Right Dataset: Hardness Criteria for Feature Selection Benchmarking," *Knowledge-Based Systems* **334**, 115022 (2026).

doi.org/10.1016/j.knosys.2025.115022 — [ScienceDirect](#)

Code & datasets: github.com/Itamarelmakia/FeatureSelect-Benchmark-Hub

Chapters 1–2 (utility prediction, resource-aware evaluation): in preparation.

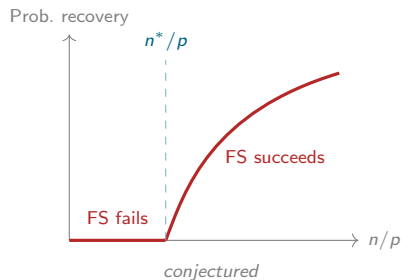
Backup: the recovery collapse vs. the PCA transition

BBP / spiked PCA: below $\lambda_c = \sqrt{p/n}$, signal recovery is information-theoretically impossible.

Peeling collapse: below a critical n , FS algorithms fail to recover F , yet the signal still exists (F works on the test set).

Difference:

- BBP: information-theoretic impossibility
- Peeling: algorithmic failure (signal present)



Backup: the 27 / 37 FS algorithms

Paper 1 (27 algorithms):

- Filter: Low Var, Skewness, Univariate, mRMR, ReliefF, LS, LeadingEV, CT, DT
- Wrapper: SVM-B/F, DecisionT-B/F, Alpha Inv
- Embedded: RFS, SVM-RBF, GRAPH, MCFS, CAE, DLFS, UFS
- Ensemble: ETree, AdaBoost
- Hybrid: CEI-GA, GRACES, SHAP, SVC

Chapters 1–2 (37 algorithms): same 27 plus SA-FS, RFE, Filter-Wrapper, Boruta, Stability-LASSO, STG, LSPIN, HSIC-Lasso, PSO-FS, Fisher Score.

Evaluation: 5-fold stratified CV;
 $K \in \{1, 5, 30, 60\}$; classifiers LR, SVM, RF, ET, XGBoost.

Backup: the 7 FS-beneficial datasets

Dataset	Domain	p	n	No-FS AUC	Δ AUC
colon	Gene	2000	62	51%	+25.3%
Period Changer	Biological	1177	90	52%	+19.3%
CNS	Gene	7129	60	58%	+16.3%
tumors_C	Gene	7129	60	58%	+16.3%
ALLAML	Gene	7129	72	89%	+8.2%
madelon	Other	500	2600	67%	+6.0%
Musk2	Chemical	166	6598	87%	+5.1%

Common pattern: very small n with large p (gene expression), or a designed synthetic challenge (madelon).

Backup: Wilcoxon tests and Cliff's δ

Comparison	Setting	Cliff's δ (Easy)	Cliff's δ (Medium)
Fast vs Slow	All	0.72 (fast)	0.60 (fast)
Library vs Research	All	0.36 (library)	0.64 (library)
Supervised vs Unsup.	All	0.32 (sup.)	0.19 (sup.)
Filter vs Hybrid	Best	0.44 (filter)	—
All vs Best	—	$\approx$ 7–12 AUC points difference	

Paired within-dataset one-sided Wilcoxon signed-rank tests. $|\delta| > 0.5$ is a large effect (Romano 2006).

Backup: hardness categorization rules (Paper 1)

Category	AUC without FS	Best AUC with FS
Easy	Any	$\geq 90\%$
Medium	Low or Fair	70–90%
Hard	Low	$< 70\%$
Fragile	High or Fair	Drops by ≥ 1 category

Performance tiers:

- High: $\text{AUC} \geq 90\%$
- Fair: 70–90%
- Low: $\text{AUC} < 70\%$

Category is dynamic: new algorithms can only raise it. Fragile and Hard mark open challenges.

Backup: references

- M. J. Post, P. van der Putten, J. N. van Rijn. “Does Feature Selection Improve Classification? A Large Scale Experiment in OpenML.” *Advances in Intelligent Data Analysis XV* (IDA 2016), LNCS 9897, 158–170. doi:10.1007/978-3-319-46349-0_14.
- A. Bommert, X. Sun, B. Bischl, J. Rahnenführer, M. Lang. “Benchmark for filter methods for feature selection in high-dimensional classification data.” *Comput. Statist. Data Anal.* 143:106839, 2020. doi:10.1016/j.csda.2019.106839.
- J. Li, K. Cheng, S. Wang, F. Morstatter, R. P. Trevino, J. Tang, H. Liu. “Feature Selection: A Data Perspective.” *ACM Comput. Surv.* 50(6):94, 2017. doi:10.1145/3136625.
- A. Filchenkov, A. Pendryak. “Datasets meta-feature description for recommending feature selection algorithm.” *AINL-ISMW FRUCT*, 11–18, 2015. doi:10.1109/AINL-ISMW-FRUCT.2015.7382962.
- A. R. S. Parmezan, H. D. Lee, F. C. Wu. “Metalearning for choosing feature selection algorithms in data mining.” *Expert Syst. Appl.* 75:1–24, 2017. doi:10.1016/j.eswa.2017.01.013.
- J. Komorniczak, P. Ksieniewicz. “proplexity — An open-source Python library for supervised data complexity assessment.” *Neurocomputing* 521:126–136, 2023. doi:10.1016/j.neucom.2022.11.056.
- I. Elmakias, D. Vilenchik. “Choosing the Right Dataset: Hardness Criteria for Feature Selection Benchmarking.” *Knowledge-Based Systems* 334:115022, 2026. doi:10.1016/j.knosys.2025.115022.